CODA: Calibrate Once, Deploy Anywhere—Reliable False-Alarm Control for Quantized Anomaly Detectors on Heterogeneous Microcontrollers

Abdelaziz Saud
PhD Program: Computer Science
University of Sharjah
Student ID: U25200513 Email: U25200513@sharjah.ac.ae
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Abstract—Anomaly detectors for cyber-physical systems are calibrated at full precision and then quantized to run on microcontrollers. Quantization shifts the anomaly-score distribution while largely preserving its ranking, so a conformal threshold inherited from the full-precision model can silently lose its false-alarm guarantee. We propose CODA (Calibrate Once, Deploy Anywhere), which calibrates the threshold once on a host GPU and gives each device a label-free correction measured from how its integer arithmetic perturbs the score. We will test whether this keeps the realized false-alarm rate within a target bound on three microcontroller families (STM32, ESP32, RP2350) using public industrial anomaly-detection benchmarks, and report latency and memory overhead on each board.

Index Terms—conformal prediction, anomaly detection, quantization, TinyML, cyber-physical systems

I. Problem Statement and Motivation

Industrial sensors increasingly detect faults on the device itself, which cuts latency and keeps raw data local. To fit in kilobytes of SRAM, models are converted to 8-bit integers (INT8) after training [1]–[3]. Operators also need a predictable false-alarm rate. Conformal anomaly detection provides one: a threshold that bounds false alarms on normal data at a chosen level α [4], [5]. That guarantee holds only for the model that was calibrated.

In practice, the threshold is set once at full precision and copied to every device. Each device runs a different integer model, with its own kernels and rounding, so its scores drift and the guarantee is lost. Recalibrating on every device is often impractical: calibration data may be too large for flash, too sensitive to ship, or too costly to collect for each hardware variant in a fleet.

Research question: Can a threshold calibrated once at full precision, plus a label-free per-device shift correction, keep the realized false-alarm rate of INT8 anomaly detectors on heterogeneous microcontrollers within a target bound?

Hypothesis: On all three boards, the corrected threshold keeps the false-alarm rate on normal test data within $\alpha+\delta$ (with $\delta = 0.01$, for $\alpha \in \{0.01, 0.05\}$), while the inherited threshold exceeds it, and detection recall stays within 5 percentage points of per-device recalibration.

TABLE I
Closest work and what it leaves open

Work	Contribution	Open
VLA monitors [10]	Inherited threshold fails under quantization	No guarantee; simulated; not sensors
LLM benchmark [11]	Conformal on compressed LLMs	Recalibrates each model
Retrieval [12]	Certifies FP/INT agreement	Not false-alarm control
TCUQ, HDUQ [13], [14]	On-MCU conformal	No FP-to-INT transfer

II. Related Work and Research Gap

Table I summarizes the closest work. A NeurIPS 2026 workshop study showed that 4-bit quantization of robot-policy failure detectors leaves AUC almost unchanged while the coverage of an inherited conformal threshold drops from 0.77 to 0.41 (target 0.90), because quantization roughly translates the score distribution [10]; its repair is empirical and its quantization simulated. A 2026 benchmark measured conformal prediction on compressed LLMs, but recalibrates every compressed model [11]. A retrieval study certifies when quantized outputs match full-precision ones, not false-alarm rates [12]. On microcontrollers, TCUQ and HDUQ-HAR run conformal calibration on the deployed model directly [13], [14]. Meanwhile, INT8 kernel paths have been shown to diverge layer by layer [15], so each chip may need its own correction. Conformal theory for non-exchangeable data exists [6], [7] but has not been applied to this setting.

Research gap. No existing method restores a conformal false-alarm guarantee for a full-precision threshold deployed on integer hardware, with proof, on real microcontrollers, for streaming sensor data.

III. Proposed Method

A. Problem Formulation

Let x be a sensor window, $s(x)$ the full-precision anomaly score, and $s_d(x)$ the score of the INT8 model on

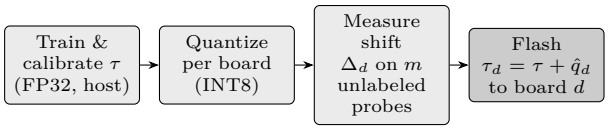


Fig. 1. The CODA pipeline. Only one number, τ_d, is sent to each board.

device d. Split-conformal calibration on n normal windows sets τ as the $\lceil (n+1)(1-\alpha) \rceil$-th smallest calibration score [4]. With score shift $\Delta_d(x) = s_d(x) - s(x)$, we set

$$\tau_d = \tau + \hat{q}_d, \tag{1}$$

where $\hat{q}_d$ is the $\lceil (m+1)(1-\delta) \rceil$-th smallest of Δ_d over m unlabeled probe windows. A union bound gives a false-alarm rate on device d of at most $\alpha + \delta$ when probe and test windows are exchangeable; we will prove this formally and extend it to temporally dependent data using block calibration [6].

B. Approach

Fig. 1 shows the pipeline. (1) Windows of length 64 (stride 1) are z-normalized with training statistics. (2) A 1D-convolutional autoencoder (three conv layers, under 100 KB in INT8) is trained on normal windows; the score is reconstruction mean-squared error. The last 20% of normal training data, split in time, is the calibration set. (3) Each board's model is produced by TensorFlow Lite full-integer post-training quantization with 500 representative windows, and compiled with TFLite Micro using CMSIS-NN (STM32, RP2350) or ESP-NN (ESP32) [2]. (4) $m = 500$ probe windows are streamed to each board over UART; on-device scores give Δ_d. (5) τ_d is flashed to the board. No calibration data leaves the host.

IV. Evaluation Plan

A. Datasets and Baselines

Data: Server Machine Dataset (SMD) [8] as the primary benchmark; SWaT (water treatment, by request) as the industrial case; NASA SMAP/MSL as a backup. Because common anomaly benchmarks have known flaws [9], the main metric is false-alarm rate on normal data, where labels are least disputed. Hardware: STM32 Nucleo (Cortex-M4/M7), ESP32-S3 (Xtensa LX7), and Raspberry Pi Pico 2 (RP2350, Cortex-M33), under 100 USD in total, plus existing lab GPUs. Baselines: (B1) inherited threshold τ; (B2) per-device recalibration with the full calibration set (upper bound); (B3) on-device streaming quantile recalibration, in the spirit of TCUQ and adaptive conformal inference [7], [13]; (B4) the label-free realignment repair of [10].

B. Metrics

Task metrics: realized false-alarm rate versus target $\alpha \in \{0.01, 0.05, 0.1\}$, detection recall, and AUC. Systems measures: inference latency per window (on-chip cycle counter), peak SRAM, and flash, including the overhead of the correction. All results are averaged over 5 seeds (training and calibration split), with 95% confidence intervals.

TABLE II
Planned experiments

ID	Question	Variants	Output
E1	Does inheriting τ break?	3 boards, 3 datasets	FAR vs. α, AUC
E2	Does τ_d hold?	Ours vs. B1–B4	FAR, recall, latency, SRAM
E3	Is the shift numerical?	Probes: in-domain, other dataset, synthetic; $m \in \{50, 200, 500\}$	FAR vs. probe source, m
E4	Robust to compression?	INT8 vs. INT4; 3 model sizes	FAR, recall, memory

TABLE III
Risk register

Risk	Level	Mitigation
Technical: shift depends on data (H2 fails)	M	Keep a small in-domain probe set on the host; report how much is needed
Technical: bound too loose	M	Tighter joint quantile instead of union bound
Novelty: "why not recalibrate per device?"	M	Quantify data size, privacy, and fleet cost; report B2 as upper bound
Data: SWaT access delayed	M	Start with SMD and SMAP/MSL
Compute: board or toolchain issues	L	Two boards suffice; drop RP2350 first
Security: tampered probe set inflates τ_d	L	Hash and sign probe set; report sensitivity
Ethics: data use	L	Public, non-personal datasets under their licenses

C. Ablations and Comparisons

Table II lists the planned experiments. E1 tests the failure; E2 tests the hypothesis against B1–B4; E3 and E4 are ablations on probe source and size, and on bit-width and model size.

V. Risk Analysis

Table III lists the main risks.

VI. Timeline and Deliverables

Week 6: proposal. Weeks 7–10: data pipeline, training and calibration, quantization for three boards, baselines B1–B2; Progress Report 1 (Week 10). Weeks 11–13: proof of the bound, on-device runs, E1–E3, baselines B3–B4; Progress Report 2 (Week 13). Weeks 14–16: E4, time-series extension, 6–8 page IEEE paper, artifact, and defense (Week 16).

VII. Reproducibility and Ethics

Repository: https://github.com/as4193/CODA, containing training, quantization, firmware, and evaluation code. Seeds 0–4 are fixed for training and splits. Data versions: SMD from the OmniAnomaly release, SWaT by its release date, with checksums recorded. Environment: Python, PyTorch, TensorFlow/TFLite Micro, STM32CubeIDE, and ESP-IDF versions pinned in an environment file and Dockerfile. Ethical, privacy, or security notes: only public, non-personal industrial data are used; the probe-set tampering risk is analyzed in Table III.

AI Assistance Disclosure

Claude (Anthropic) was used to brainstorm project directions, search and summarize recent literature, and draft the proposal text and LaTeX, and to look up author lists for recent references. I verified the title, authors, and venue of each reference against the original paper or its arXiv/publisher page, and checked the method, hypothesis, and experiment plan against the course guidelines. I reviewed and edited the final text and take responsibility for its content.

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